



SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (AUTONOMOUS)

(Approved by AICTE, New Delhi, Affiliated to JNTUK, Kakinada)

Accredited by NAAC with 'A+' Grade.

Recognised as Scientific and Industrial Research Organisation

SRKR MARG, CHINA AMIRAM, BHIMAVARAM – 534204 W.G.Dt., A.P., INDIA

Regulation: R23			IV / IV - B.Tech. I - Semester						
CSE (IoT and Cyber Security including Blockchain Technology)									
COURSE STRUCTURE (With effect from 2023-24 admitted Batch onwards)									
Course Code	Course Name	Category	L	T	P	Cr	C.I.E.	S.E.E.	Total Marks
B23CI4101	Ad-hoc Sensor Networks	PC	3	0	0	3	30	70	100
B23HS4102	Production &Operations Management	HS	2	0	0	2	30	70	100
#PE-IV	Professional Elective-IV	PE	3	0	0	3	30	70	100
#PE-V	Professional Elective-V	PE	3	0	0	3	30	70	100
#OE-III	Open Elective-III	OE	3	0	0	3	30	70	100
#OE-IV	Open Elective-IV	OE	3	0	0	3	30	70	100
B23CI4113	Ethical Hacking	SEC	0	1	2	2	30	70	100
B23MC4101	Constitution of India	MC	2	0	0	0	30	-	30
B23CI4114	Evaluation of Industry Internship /Mini Project	PR	0	0	0	2	-	50	50
TOTAL			19	1	2	21	240	540	780

	Course Code	Course
#PE-IV	B23CI4102	IoT Applications
	B23CI4103	Malware Analysis & Reverse Engineering
	B23CI4104	Crypto Currency Technologies
	B23CI4105	Design of IoT Architecture
	B23CI4106	12-week MOOC Swayam/NPTEL course recommended by the BoS
#PE-V	B23CI4107	Cyber Laws and Security Policies
	B23CI4108	Intrusion Detection and Prevention System
	B23CI4109	Metaverse
	B23CI4110	Smart Contract Essentials
	B23CI4111	Augmented Reality & Virtual Reality
	B23CI4112	12-week MOOC Swayam/NPTEL course recommended by the BoS
#OE-III	Student has to study one Open Elective offered by CE or ME or ECE or EEE from the list enclosed.	
#OE-IV	Student has to study one Open Elective offered by CE or ME or ECE or EEE from the list enclosed.	

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23CI4101	PC	3	--	--	3	30	70	3 Hrs.
AD-HOC SENSOR NETWORKS								
(For CIC)								
Course Objectives:								
1.	To understand the concepts of sensor networks							
2.	To understand the MAC and transport protocols for ad hoc networks							
3.	To understand the security of sensor networks							
4.	To understand the applications of adhoc and sensor networks							
Course Outcomes: At the end of the Course, students will be able to								
S.No	Outcome							Knowledge Level
1.	Explain the characteristics, applications, challenges, and routing approaches of Mobile Ad Hoc Networks							K2
2.	Analyze broadcast control mechanisms and multicasting protocols used for data transmission in MANETs.							K3
3.	Evaluate geo casting mechanisms and TCP performance issues in MANET environments.							K4
4.	Explain the architecture and lower-layer protocols of Wireless Sensor Networks including physical, MAC, link, and routing layers.							K2
5.	Analyze upper-layer issues in Wireless Sensor Networks, including transport protocols, application layer services, and adaptation to dynamic network environments.							K3
SYLLABUS								
UNIT-I (8 Hrs)	Introduction to Ad Hoc Networks - Characteristics of MANETs, Applications of MANETs and Challenges of MANETs. Routing in MANETs - Criteria for classification, Taxonomy of MANET routing algorithms, Topology based routing algorithms, Proactive: DSDV, Reactive: DSR, AODV, Hybrid: ZRP, Position-based routing algorithms, Location Services, DREAM, Quorum-based, Forwarding Strategies: Greedy Packet, Restricted Directional Flooding-DREAM, LAR.							
UNIT-II (8 Hrs)	Data Transmission - Broadcast Storm Problem, Rebroadcasting Schemes-Simple-flooding, Probability-based Methods, Area-based Methods, Neighbor Knowledge-based: SBA, Multipoint Relaying, AHBP. Multicasting: Tree-based: AMRIS, MAODV, Mesh-based: ODMRP, CAMP, Hybrid: AMRoute, MCEDAR.							

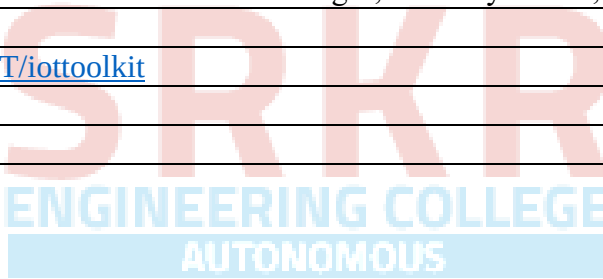
UNIT-III (12 Hrs)	Geocasting: Data-transmission Oriented-LBM, Route Creation Oriented-GeoTORA, MGR. TC over Ad Hoc TCP protocol overview, TCP and MANETs, Solutions for TCP over Ad hoc
UNIT-IV (10Hrs)	Basics of Wireless, Sensors and Lower Layer Issues: Applications, Classification of sensor networks, Architecture of sensor network, Physical layer, MAC layer, Link layer, Routing Layer.
UNIT-V (10Hrs)	Upper Layer Issues of WSN: Transport layer, High-level application layer support, Adapting to the inherent dynamic nature of WSNs, Sensor Networks and mobile robots.
Textbooks:	
1.	Ad Hoc and Sensor Networks – Theory and Applications, Carlos Corderio Dharma P. Aggarwal, World Scientific Publications, March 2006, ISBN – 981–256–681–3.
2.	Wireless Sensor Networks: An Information Processing Approach, Feng Zhao, Leonidas Guibas, Elsevier Science, ISBN – 978-1-55860-914-3 (Morgan Kauffman).
Reference Books:	
1.	Carlos Cordeiro and Dharma P. Agrawal, “Ad Hoc and Sensor Networks: Theory and Applications.” World Scientific Publishing, 2006.
2.	C. Siva Ram Murthy and B. S. Manoj, “Ad Hoc Wireless Networks: Architectures and Protocols.” Pearson Education, 2004.
e-Resources	
1.	https://nptel.ac.in/courses/112107238
2.	https://alison.com/course/introduction-to-connectivity-technologies-and-sensor-networks

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23HS4102	HS	2	--	--	2	30	70	3 Hrs.
PRODUCTION & OPERATIONS MANAGEMENT								
(For CIC & CSBS)								
Course Objectives:								
1.	To impart a comprehensive understanding of Production &Operations Management principles and practices.							
2.	To equip students with knowledge of product and process design principles, innovation, development, and optimization techniques.							
3.	To provide knowledge of supply chain strategies, inventory management, logistics, demand forecasting, and operational efficiency							
4.	To develop understanding of quality management principles, continuous improvement methods, process optimization, and performance enhancement strategies.							
5.	To equip students with knowledge of operations planning, scheduling, control techniques, resource management, and process optimization.							
Course Outcomes: At the end of the Course, students will be able to								
S.No	Outcome							Knowledge Level
1.	Apply the principles of Production &Operations Management in analyzing production processes and operational activities in organizations							K3
2.	Apply the principles of product and process design, innovation, optimization, sustainability, and manufacturing efficiency.							K3
3.	Implement qualitative and quantitative demand forecasting techniques for supply chain planning.							K3
4.	Illustrate the importance of global quality standards such as ISO series for organizational compliance.							K3
5.	Implement strategic production planning and operational scheduling to improve organizational productivity.							K3
SYLLABUS								
UNIT-I (8 Hrs)	Introduction to Production &Operations Management: Definition, Nature, and Scope of Operations, Variables influencing the Production & Operations Management Management : Role and Significance; Evolution of Operations Management: Historical Perspectives; Operations Strategy: Formulation and Implementation							
UNIT-II (8 Hrs)	Product and Process Design: Product Development Life Cycle and Innovation Strategies; Production System Classifications: Job, Batch, Mass, and Continuous Production; Facility Design: Floor Plan Mapping & types; Location Optimization & Site Selection Criteria							
UNIT-III	Supply Chain and Inventory Management: Overview of Supply Chain Management:							

(12 Hrs)	Concepts and Importance; Inventory Control Systems: Economic Order Quantity, ABC Analysis, Just In Time; Demand Forecasting Techniques: Qualitative and Quantitative Models; Digital Supply Chains: Role of Technology and Automation
UNIT-IV (10Hrs)	Quality Management and Improvement: Fundamentals of Quality Management and TQM Principles, Quality Assurance Tools: Six Sigma, Kaizen, SPC, and Benchmarking; Global Quality Standards: ISO Series and Compliance.
UNIT-V (10Hrs)	Operations Planning and Control: Strategic Production Planning and Operational Scheduling; Capacity Planning Techniques and Resource Optimization; Project Management Methodologies: Critical Path Method; Program Evaluation and Review Technique
Textbooks:	
1.	Aswathappa, K., & Bhattacharya, S. (2010). Production and Operations Management. Himalaya Publishing House
2.	Panneerselvam, R. (2012). Production and Operations Management. PHI Learning
3.	KanishkaBedi (2018). Production and Operations Management. Viva Books.
Reference Books:	
1.	Heizer, J., Render, B., & Munson, C. (2017). Operations Management. Pearson Education.
2.	Stevenson, W. J. (2020). Operations Management. McGraw-Hill Education.
3.	Chase, R. B., Jacobs, F. R., & Aquilano, N. J. (2019). Operations and Supply Chain Management. McGraw-Hill.
4.	Slack, N., & Brandon-Jones, A. (2019). Operations Management. Pearson.
5.	Krajewski, L. J., Malhotra, M. K., & Ritzman, L. P. (2018). Operations Management: Processes and Supply Chains. Pearson Education.
e-Resources	
1.	https://nptel.ac.in/courses/112107238

Course Code	Category	L	T	P	C	I.M	E.M	Exam
B23CI4102	PE	3	--	--	3	30	70	3 Hrs.
IoT Applications								
(For CIC)								
Course Objectives: Students are expected								
1.	Interpret the key components, enabling technologies and infrastructure supporting IoT systems.							
2.	Outline the role of M2M communication and industrial value chains in IoT ecosystems.							
3.	Interpret IoT reference architectures and architectural views used in system development.							
4.	Investigate the implementation of IoT solutions in industrial, healthcare and smart environment domains.							
5.	Investigate security, privacy and governance aspects associated with IoT platforms and smart city systems.							
Course Out Comes: At the end of the course students will be able to								
S. No	OUT COME							KL
1.	Describe IoT concepts, technologies, infrastructure and communication processes.							K2
2.	Apply M2M communication and IoT value chain concepts in IoT systems.							K3
3.	Apply IoT architectural models and reference architectures for system design.							K3
4.	Analyze IoT applications in industrial, healthcare, home and smart environments.							K4
5.	Analyze IoT security, privacy and governance issues in smart systems.							K4
SYLLABUS								
UNIT-I (10 Hrs)	IoT & Web Technology, The Internet of Things Today, Time for Convergence, Towards the IoT Universe, Internet of Things Vision, IoT Strategic Research and Innovation Directions, IoT Applications, Future Internet Technologies, Infrastructure, Networks and Communication, Processes, Data Management, Security, Privacy & Trust, Device Level Energy Issues, IoT Related Standardization, Recommendations on Research Topics.							
UNIT-II (10 Hrs)	M2M to IoT – A Basic Perspective – Introduction, Some Definitions, M2M Value Chains, IoT Value Chains, An emerging industrial structure for IoT, The international driven global value chain and global information monopolies. M2M to IoT – An Architectural Overview – Building an architecture, Main design principles and needed capabilities, An IoT architecture outline, standards considerations.							
UNIT-III (10 Hrs)	IoT Architecture – State of the Art – Introduction, State of the art, Architecture. Reference Model – Introduction, Reference Model and architecture, IoT reference model, IoT Reference Architecture – Introduction, Functional View, Information View, Deployment and Operational View, Other Relevant architectural views.							
UNIT-IV	IoT Applications for Value Creations – Introduction, IoT applications for industry:							

(10 Hrs)	Future Factory Concepts, Brownfield IoT, Smart Objects, Smart Applications, Four Aspects in your Business to Master IoT, Value Creation from Big Data and Serialization. IoT for Retailing Industry, IoT For Oil and Gas Industry, Opinions on IoT Application and Value for Industry, Home Management, eHealth.
UNIT-V (10 Hrs)	Internet of Things Privacy, Security and Governance – Introduction, Overview of Governance, Privacy and Security Issues, Contribution from FP7 Projects, Security, Privacy and Trust in IoT Data Platforms for Smart Cities, First Steps Towards a Secure Platform, Smartie Approach. Data Aggregation for the IoT in Smart Cities, Security.
TEXTBOOK:	
1.	Vijay Madiseti and Arshdeep Bahga, “Internet of Things: (A Hands-on Approach)”, Universities Press (INDIA) Private Limited, 2014, 1st Edition.
REFERENCE BOOKS:	
1.	Michael Miller, “The Internet of Things: How Smart TVs, Smart Cars, Smart Homes, and Smart Cities Are Changing the World”, Pearson Education, 2015.
2.	Francis da Costa, “Rethinking the Internet of Things: A Scalable Approach to Connecting Everything”, Apress Publications, 2013, 1st Edition.
3.	Waltenegus Dargie, Christian Poellabauer, “Fundamentals of Wireless Sensor Networks: Theory and Practice”, Wiley, 2014.
4.	Cuno Pfister, “Getting Started with the Internet of Things”, O’Reilly Media, 2011.
E-Resources:	
1.	https://github.com/connectIOT/iottoolkit
2.	https://www.arduino.cc/
3.	https://www.zettajs.org/



Course Code	Category	L	T	P	C	I.M	E.M	Exam
B23CI4103	PE	3	--	--	3	30	70	3 Hrs.
MALWARE ANALYSIS & REVERSE ENGINEERING								
(For CIC)								
Pre-requisites: OS, Networking, Security Principles								
Course Objectives: Students are expected								
1.	To learn fundamentals of malware analysis which includes analysis of JIT compilers for malware detection in legitimate code							
2.	To explore the techniques for detecting, analyzing, reverse engineering and eradicating malware.							
3.	Employ network and system-monitoring tools to examine how malware interacts with the file system, registry, network, and other processes in a Windows environment.							
4.	Assess the threat associated with malicious documents.							
Course Outcomes: At the end of the course students will be able to								
S. No	OUT COME							KL
1.	Apply the concept of malware and reverse engineering to create Custom Clam AV Databases							K3
2.	Apply various tools and techniques of malware analysis to examine malicious software.							K3
3.	Use various debugging techniques for malware analysis to identify vulnerabilities.							K3
4.	Investigate the process of memory forensics to identify injected code.							K3
5.	Use WHOIS, DNS for creating reverse IP, static and interactive maps.							K3
SYLLABUS								
UNIT-I (10 Hrs)	Fundamentals of Malware Analysis (MA), Reverse Engineering Malware (REM) Methodology, Brief Overview of Malware analysis lab setup and configuration, Introduction to key MA tools and techniques, Behavioural Analysis vs. Code Analysis, Resources for Reverse-Engineering Malware (REM) Understanding Malware Threats, Malware indicators, Malware Classification, Examining Clam AV Signatures, Creating Custom Clam AV Databases							
UNIT-II (10 Hrs)	Malware Forensics Using TSK for Network and Host Discoveries, Using Microsoft Offline API to Registry Discoveries, Identifying Packers using PEiD, Registry Forensics with Reg Ripper Plu-gins, Bypassing Poison Ivy’s Locked Files, Bypassing Conficker’s File System ACL Restrictions, Detecting Rogue PKI Certificates.							
UNIT-III (10 Hrs)	Malware and Kernel Debugging Opening and Attaching to Processes, Configuration of JITDebugger for Shellcode Analysis, Controlling Program Execution, Setting and Catching Breakpoints, Debugging with Python Scripts and Py Commands, DLL Export Enumeration, Execution, and Debugging, Debugging a VMware Workstation Guest (on							

	Windows), Debugging a Parallels Guest (on Mac OS X).
UNIT-IV (10 Hrs)	Memory Forensics and Volatility Memory Dumping with MoonSols Windows Memory Toolkit, Accessing VM Memory Files Overview of Volatility, Investigating Processes in Memory Dumps, Code Injection and Extraction, Detecting and Capturing Suspicious Loaded DLLs, Finding Artifacts in Process Memory, Identifying Injected Code with Malfind and YARA.
UNIT-V (8 Hrs)	Researching and Mapping Source Domains/IPs Using WHOIS to Research Domains, DNS Hostname Resolution, Querying Passive DNS, Checking DNS Records, Reverse IP Search New Course Form, Creating Static Maps, Creating Interactive Maps.
TEXTBOOK:	
1.	Sikorski, M., & Honig, "A. Practical malware analysis: the hands-on guide to dissecting malicious software". No starch press,2012
2.	Malin, C. H. (2013). Linux Malware Incident Response: A Practitioner's Guide to Forensic Collection and Examination of Volatile Data: An Excerpt from Malware Forensic Field Guide for Linux Systems. Elsevier.
REFERENCE BOOKS:	
1.	Shashidhar, N., & Cooper, P. (2016, April). Teaching malware analysis: The design philosophy of a model curriculum. In 2016 4th International Symposium on Digital Forensic and Security (ISDFS) (pp. 119-125). IEEE.
E-Resources:	
1.	https://www.infosecinstitute.com/resources/malware-analysis/malware-analysis-and-reverse-engineering/

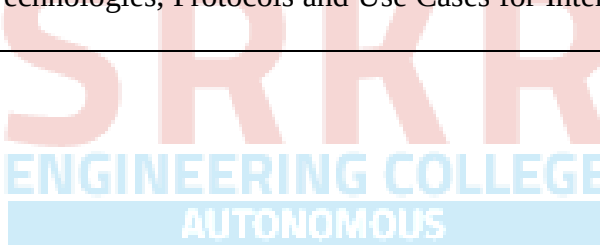
Course Code	Category	L	T	P	C	I.M	E.M	Exam
B23CI4104	PE	3	--	--	3	30	70	3 Hrs.
CRYPTO CURRENCY TECHNOLOGIES								
(For CIC)								
Pre-requisites: Cryptography and Network Security.								
Course Objectives: Students are expected								
1.	To understand the importance of digital money through crypto mechanisms							
2.	To be able to explain the different components involved within Blockchain							
3.	To understand how the block chain creates economic incentives							
4.	To know the system works and how can they utilize and what application can be build using cryptocurrency							
Course Out Comes: At the end of the course students will be able to								
S.No	OUT COME							KL
1.	Identify how cryptographic primitives can be used to construct secure electronic currencies like bit coin.							K3
2.	Illustrate emerging abstract models for Blockchain Technology							K2
3.	Demonstrate the Bitcoin features and its alternative options.							K2
4.	Develop smart contracts on Ethereum using solidity language.							K3
5.	Build Block Chain Applications using the learned concepts.							K3
SYLLABUS								
UNIT-I (10 Hrs)	DIGITAL MONEY: Physical Money and Digital Money, History of Money, Forms of Money, Fiat Currencies, Currency Pegs, Quantitative Easing, Interbank Payment Mode: Same Bank, Different Bank, Correspondent Bank, Central Bank, International Payment. CRYPTOGRAPHY: Encryption and Decryption, Hashes, Digital Signature, ECDSA, Memory Hard Algorithm, Zero Knowledge Proof.							
UNIT-II (10 Hrs)	BLOCKCHAIN TECHNOLOGY: Blockchain Introduction, Advantage over conventional distributed database, Blockchain Network, Mining Mechanism, Distributed Consensus, Merkle Patricia Tree, Gas Limit, Transactions and Fee, Anonymity, Reward, Chain.							
UNIT-III (10 Hrs)	CRYPTOCURRENCY AND BLOCKCHAIN TECHNOLGOY: Introduction to crypto and cryptocurrency, Primitives of crypto, Bitcoin, Point of Bitcoin, Working of Bitcoin, Bitcoin in practice, Bitcoin Predecessor, Bitcoin Decentralization, Store and use of Bitcoin, Bit coin Mining. Ethereum: Ethereum Eco system. Case Study: Bitcoin cash.							
UNIT-IV (10 Hrs)	ETHEREUM AND SOLIDITY: Alternative approaches to mining and consensus, Bitcoin and Anonymity, Overview of Altcoins, Overview of Ethereum, Programming smart contracts on Ethereum, Solidity Language. Crypto currencies & the real world: Smart							

	property, data feeds, Applications of cryptocurrencies and blockchains.
UNIT-V (10 Hrs)	CRYPTOCURRENCY REGULATION AND BLOCKCHAIN APPLICATIONS: Crypto currency Regulation: Stakeholders, Roots of Bitcoin, Legal Aspects, Crypto currency Exchange, Black Market and Global Economy. Blockchain Applications: Internet of Things, Medical Record Management System, Domain Name Service, Future of Blockchain.
TEXTBOOK:	
1.	Antony Lewis, "The Basics of Bitcoin and Block chains: An Introduction to Cryptocurrencies and Technology that powers them"2018.
2.	Wiley, "Crypto currencies and Block chain Technology Applications", Wiley2020.
REFERENCE BOOKS:	
1.	Arvind Narayanan, Joseph Bonneau, Edward Felten, Andrew Miller and Steven Goldfeder, "Bit coin and Crypto currency Technologies: A Comprehensive Introduction ", Princeton University Press (July19, 2016).
2.	Antono poulos, "Mastering Bit coin: Unlocking Digital Crypto currencies", O'Reilly(2014)



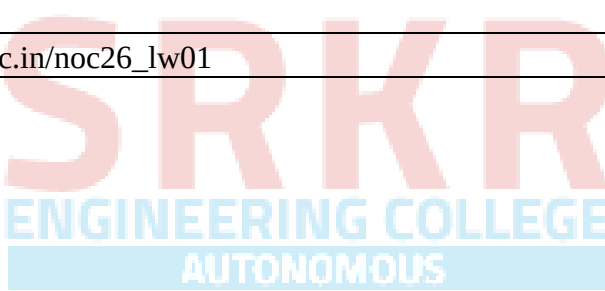
Course Code	Category	L	T	P	C	I.M	E.M	Exam
B23CI4105	PE	3	--	--	3	30	70	3 Hrs.
DESIGN OF IOT ARCHITECTURE								
(For CIC)								
Course Objectives: Students are expected								
1.	Explain the IoT ecosystem, applications and architectural models used in modern IoT systems.							
2.	Describe IoT devices, wireless networks and protocol concepts used in IoT communication.							
3.	Illustrate IoT device design considerations and event-driven system analysis.							
4.	Explain Industrial IoT (IIoT) architecture and Industry 4.0 technologies.							
5.	Describe security, safety mechanisms and testing techniques used in IoT systems							
Course Outcomes: At the end of the course students will be able to								
S. No	OUT COME							KL
1.	Explain IoT applications, architectures, wireless networks and device ecosystem.							K2
2.	Interpret IoT device design space, network models and event-driven system analysis.							K3
3.	Apply the concepts of Industrial IoT architecture and enabling technologies to industrial scenarios.							K3
4.	Examine security, safety, privacy and dependable design issues in IoT systems.							K4
5.	Analyze fuzz testing methods and security testing approaches used in IoT systems.							K4
SYLLABUS								
UNIT-I (10 Hrs)	The IoT Landscape: What is IoT?, Applications, Architectures, Wireless Networks, Devices, Security and Privacy, Event-Driven Systems. IoT System Architectures: Introduction, Protocol Concepts, IoT-Oriented Protocols, Databases, Time Bases, Security.							
UNIT-II (10 Hrs)	IoT Devices & Event-Driven System Analysis: The IoT Device Design Space, Cost of Ownership and Power Consumption, Cost per Transistor and Chip Size, Duty Cycle and Power Consumption, Platform Design. Event-Driven System Analysis: Introduction, Motivating Example, IoT Network Model, Events, Networks, Devices and Hubs, Single-Hub Networks, Multi-Hub Networks, Network Models and Physical Networks, IoT Event Analysis, Event Populations, Stochastic Event Populations, Environmental Interaction Modeling, Event Transport and Migration.							
UNIT-III (10 Hrs)	Industrial Internet of Things: Introduction, Industry 4.0, Industrial Internet of Things (IIoT), IIoT Architecture, Basic Technologies, Applications and Challenges.							

UNIT-IV (10 Hrs)	Security and Safety: Introduction, Systems Security, Network Security, Generic Application Security, Application Process Security and Safety, Reliable-and-Secure-by-Design IoT Applications, Run-Time Monitoring, The ARMET Approach, Privacy and Dependability.
UNIT-V (10 Hrs)	Security Testing IoT Systems: Introduction, Fuzz Testing for Security, White-Box Fuzzing, Black-Box Fuzzing, Fuzzing Industrial Control Network Systems, Fuzzing Modbus, The Modbus Protocol, Modbus/TCP Fuzzer.
TEXTBOOK:	
1.	Dimitrios Serpanos, Marilyn Wolf, Internet-of-Things (IoT) Systems Architectures, Algorithms, Methodologies, ISBN 978-3-319-69714-7.
REFERENCE BOOKS:	
1.	Arshdeep Bahga, Vijay Madiseti, Internet of Things – A Hands-on Approach, Universities Press, 2015.
2.	Olivier Hersent, David Boswarthick, Omar Elloumi, The Internet of Things – Key Applications and Protocols, Wiley, 2012 (for Unit 2).
3.	Jan Holler, Vlasios Tsiatsis, Catherine Mulligan, Stamatis Karnouskos, Stefan Avesand, David Boyle, From Machine-to-Machine to the Internet of Things – Introduction to a New Age of Intelligence, Elsevier, 2014.
4.	David Hanes, Gonzalo Salgueiro, Patrick Grossetete, Rob Barton, Jerome Henry, IoT Fundamentals: Networking Technologies, Protocols and Use Cases for Internet of Things, Cisco Press, 2017.



Course Code	Category	L	T	P	C	I.M	E.M	Exam
B23CI4107	PE	3	--	--	3	30	70	3Hrs.
CYBER LAWS AND SECURITY POLICIES								
(For CIC)								
Pre-requisites: CNS, Cyber Security								
Course Objectives:								
1.	To understand cyber laws and different security policies							
2.	To know the different ethical responsibilities In the present world							
3.	To understand how the cyber employ will be in the organization							
4.	To understand the different organization and human adoption rights							
Course Outcomes: At The End Of the course students will be able								
S. No	OUT COME							KL
1.	Explain the fundamental concepts of computer security, including confidentiality, integrity, and availability.							K2
2.	Classify various types of security threats and vulnerabilities in computer systems.							K3
3.	Apply appropriate access control mechanisms and information protection techniques in computing environments.							K3
4.	Describe computer security standards, government requirements, and relevant legislation.							K2
5.	Apply appropriate methods to address privacy issues and implement international security practices in information systems.							K3
SYLLABUS								
UNIT-I (10 Hrs)	Introduction to Computer Security: Definition, Threats to security, Government requirements, Information Protection and Access Controls, Computer security efforts, Standards, Computer Security mandates and legislation, Privacy considerations, International security activity.							
UNIT-II (10 Hrs)	Secure System Planning and administration, Introduction to the orange book, Security policy requirements, accountability, assurance and documentation requirements, Network Security, The Redbook and Government network evaluations.							
UNIT-III (10 Hrs)	Information security policies and procedures: Corporate policies- Tier 1, Tier 2 and Tier3 policies - process management-planning and preparation-developing policies-asset classification policy developing standards.							
UNIT-IV	Information security: fundamentals-Employee responsibilities- information							

(10 Hrs)	classification Information handling- Tools of information security- Information processing-secure program administration.
UNIT-V (8 Hrs)	Organizational and Human Security: Adoption of Information Security Management Standards, Human Factors in Security- Role of information security professionals.
TextBooks:	
1.	Debby Russell and Sr. G.T Gangemi, "Computer Security Basics (Paperback)", 2ndEdition, O' Reilly Media, 2006.
2.	Thomas R. Peltier, "Information Security policies and procedures: A Practitioner's Reference", 2nd Edition Prentice Hall, 2004.
ReferenceBooks:	
1.	Kenneth J. Knapp, "Cyber Security and Global Information Assurance: Threat Analysis And Response Solutions", IGI Global, 2009.
2.	Thomas R Peltier, Justin Peltier and John blackley, "Information Security Fundamentals", 2ndEdition, Prentice Hall, 1996
3.	Jonathan Rosenoer, "Cyber law: the Law of the Internet", Springer-verlag, 1997
4.	James Graham, " Cyber Security Essentials" Averbach Publication T & F Group.
e-Resources	
1	https://onlinecourses.nptel.ac.in/noc26_lw01



Course Code	Category	L	T	P	C	I.M	E.M	Exam
B23CI4108	PE	3	--	--	3	30	70	3Hrs.
INTRUSION DETECTION AND PREVENTION SYSTEM								
(For CIC)								
Pre-requisites: CNS, Cyber Security								
Course Objectives:								
1.	To understand Intrusion Prevention Systems, Network IDs protocol and model for intrusion analysis.							
2.	To Apply knowledge of the fundamentals and history of Intrusion Detection in order to avoid common pitfalls in the creation and evaluation of new Intrusion Detection Systems.							
3.	To Understand when, where, how, and why to apply Intrusion Detection tools and techniques in order to improve the security posture of an enterprise.							
4.	To learn agent development for intrusion detection and architectural models of IDs and IPs.							
Course Outcomes: At The End Of the course students will be able								
1.	Explain basic concepts of intrusion detection systems.							K2
2.	Apply Intrusion Prevention Systems, Network IDs protocol and model for intrusion analysis.							K3
3.	Use snort to install and learn different alert modes.							K3
4.	Apply snort rules to configure files and mysql.							K3
5.	Use ACID, agent development for intrusion detection and learn architectural models of IDs and IPs.							K3
SYLLABUS								
UNIT-I (10 Hrs)	History of Intrusion detection, Audit, Concept and definition, Internal and external threats to data, attacks,Need and types of IDS, Information sources Host based information sources, Network based information sources.							
UNIT-II (10 Hrs)	Intrusion Prevention Systems, Network IDs protocol based IDs, Hybrid IDs, Analysis schemes, thinking about intrusion. A model for intrusion analysis, techniques Responses requirement of responses, types of responses mapping responses to policy Vulnerability analysis, credential analysis non credential analysis							
UNIT-III (10 Hrs)	Introduction to Snort, Snort Installation Scenarios, Installing Snort, Running Snort on Multiple Network Interfaces, Snort Command Line Options. Step-By-Step Procedure to Compile and Install Snort Location of Snort Files, Snort Modes Snort Alert Modes.							
UNIT-IV	Working with Snort Rules, Rule Headers, Rule Options, The Snort Configuration File							

(10 Hrs)	etc. Plugins, Preprocessors and Output Modules, Using Snort with MySQL.
UNIT-V (8 Hrs)	Using ACID and Snort Snarf with Snort, Agent development for intrusion detection, Architecture models of IDs and IPs.
Text Books:	
1.	Carl Endorf, Eugene Schultz and Jim Mellander “Intrusion Detection & Prevention”, 1st Edition, Tata McGraw-Hill, 2004.
2.	Rafeeq Rehman :“ Intrusion Detection with SNORT, Apache, MySQL, PHP and ACID,” 1st Edition, Prentice Hall , 2003.
Reference Books:	
1.	Christopher Kruegel, Fredrik Valeur, Giovanni Vigna: “Intrusion Detection and Correlation Challenges and Solutions”, 1st Edition, Springer, 2005.
2.	T. Fahringer, R. Prodan, “A Text book on Grid Application Development and Computing Environment”. 6th Edition, Khanna Publishers, 2012.
e-Resources	
1	https://nptel.ac.in/courses/106106178



Course Code	Category	L	T	P	C	I.M	E.M	Exam
B23CI4109	PE	3	--	--	3	30	70	3 Hrs.
METAVERSE								
(Common to CIC & CSBS)								
Pre-requisites: Basic computer knowledge, Internet, Networking and Programming basics								
Course Objectives: Students are expected								
1.	To understand the fundamental concepts of the Metaverse, including its architecture, technologies, and applications.							
2.	To explore Virtual Reality (VR), Augmented Reality (AR), and Mixed Reality (MR) as core components of the Metaverse.							
3.	To study blockchain, NFTs, and decentralized finance (DeFi) as economic foundations of the Metaverse.							
4.	To analyze Metaverse-based social interactions, business models, and ethical considerations							
5.	To gain hands-on experience with tools and platforms used to develop Metaverse applications.							
Course Out Comes: At the end of the course students will be able to								
S. No	OUT COME							KL
1.	Apply the concepts of Metaverse architecture and enabling technologies to develop simple use cases or prototypes for real-world applications such as education, gaming, or virtual collaboration.							K3
2.	Apply the concepts of Virtual Reality (VR), Augmented Reality (AR), and Mixed Reality (MR) to develop and demonstrate immersive applications as core components of the Metaverse.							K3
3.	Apply the concepts of Blockchain, NFTs, and Decentralized Finance (DeFi) to design and demonstrate digital economic models within the Metaverse.							K3
4.	Analyse Metaverse-based social interactions, business models, and ethical considerations to understand their impact on users and digital ecosystems.							K4
5.	Apply development tools and platforms to build and demonstrate basic Metaverse applications, gaining hands-on experience in immersive environment creation.							K3
SYLLABUS								
UNIT-I (10 Hrs)	Introduction to Metaverse: Definition and Evolution of the Metaverse, Key Components: VR, AR, MR, AI, and Blockchain, Applications of Metaverse: Gaming, Education, Healthcare, and Business. Challenges and Ethical Issues in the Metaverse							
UNIT-II (10 Hrs)	Virtual Reality (VR) and Augmented Reality (AR): Fundamentals of VR and AR Technologies, Hardware and Software Requirements Interaction Techniques and User Experience, Metaverse Platforms: Oculus, Microsoft Mesh, Horizon Worlds, Hands-on: Creating a Simple VR/AR Environment							
UNIT-III	Blockchain and Decentralization in the Metaverse: Introduction to Blockchain Technology,							

(10 Hrs)	Cryptocurrencies, NFTs, and Smart Contracts, Decentralized Applications (DApps) and Web3, Security and Privacy considerations Hands-on: Deploying an NFT on a Test Blockchain
UNIT-IV (10 Hrs)	Metaverse Economy and Social Structures: Digital Goods, Virtual Real Estate, and Digital Identity, Economic Models: Play- to-Earn, DeFi, and Token omics, Social and Ethical Aspects: Digital Citizenship and Governance, Future Trends in Metaverse Economy
UNIT-V (10 Hrs)	Tools and Development in the Metaverse: Metaverse Development Platforms: Unity, Unreal Engine, and WebXR, AI and Cloud Computing in the Metaverse, 3D Asset Creation and Interoperability Standards, Hands-on: Building a Basic Metaverse Application
Text books:	
1.	Pallavi Agarwal, Metaverse Fundamentals: Easy Hands-on Book on Understanding Metaver Buying Land, NFTs, Virtual Reality, Augmented Reality, Blockchain & Crypto Art. Notion pre Publisher.
2.	Tony Parisi, Learning Virtual Reality: Developing Immersive Experiences and Applications for desktop, Web, and Mobile, O'Reilly Media.
Reference Books:	
1.	Matthew Ball, The Metaverse: And How It Will Revolutionize Everything, Liveright Publishing.
2.	Lalit Kumar Awasthi, Sanjay Kumar, Metaverse: Concept, Content and Context , Springer 2023
3.	William R. Sherman, Understanding Virtual Reality, Morgan kaufmann-2 nd edition 2018
4.	Daniel Drescher, Blockchain Basics, Apress -2017
5.	Mark Van Rijmenam, Step into the Metaverse: How the Immersive Internet Will Unlock a Trillion-Dollar Social Economy, Wiley.
6.	Josh O’Kane, The Metaverse Economy: How Web3, Blockchain, and AI Are Shaping the Future of Business.

Course Code	Category	L	T	P	C	I.M	E.M	Exam
B23CI4110	PE	3	--	--	3	30	70	3 Hrs.
SMART CONTRACTS ESSENTIALS								
(For CIC)								
Pre-requisites: Cryptography and Network Security, Block Chain Technologies								
Course Objectives: After completion of the course, students will have adequate background, conceptual clarity and knowledge related to								
1.	Crypto currency, Bitcoin and Smart contracts							
2.	Public blockchain platforms such as Ethereum							
Course Out Comes: At the end of the course students will be able to								
S. No	OUT COME							KL
1.	Interpret the concept of smart contracts in blockchain.							K2
2.	Illustrate the Ethereum public blockchain platform							K2
3.	Interpret the need and working of crypto currency							K2
4.	Illustrate the cryptocurrency regulations							K2
5.	Demonstrate the underlying technology of Bitcoin							K2
SYLLABUS								
UNIT-I (10 Hrs)	Smart Contracts: Introduction to Smart Contracts, Structure of Smart Contract, Smart Contract Interaction, Contracts, Patterns and Smart Contracts Examples.							
UNIT-II (10 Hrs)								
Ethereum Blockchain Components: Introduction to Ethereum Development Tools, Ethereum Clients, Ethereum Languages, Ethereum Wallets, Ethereum Accounts, Ethereum Key pairs, Ethereum Platform.								
UNIT-III (10 Hrs)								
Cryptocurrency: History Distributed Ledger, Bitcoin protocols, Mining strategy and rewards, Ethereum Construction, DAO, Smart Contract, GHOST, Vulnerability, Attacks, Side chain, Name coin.								
UNIT-IV (10 Hrs)								
Cryptocurrency Regulation: Stakeholders, Roots of Bitcoin, Legal Aspects, Cryptocurrency Exchange, Black Market and Global Economy. Applications: Internet of Things, Medical Record Management System, Domain Name Service and future of Blockchain for cryptocurrencies.								
UNIT-V (10 Hrs)								
Bitcoins: Introduction to Bitcoins, Bitcoin: Digital Signature, Digital Keys, Private Keys, Public Keys, Bitcoins Addresses, Bitcoins Transactions, Bitcoins Network, Bitcoins Wallets, Bitcoins Payments, Bitcoins Clients and APIs, Bitcoins Limitations.								
TEXTBOOK:								
1.	Arvind Narayanan, Joseph Bonneau, Edward Felten, Andrew Miller and Steven Goldfeder,							

	“Bitcoin and Cryptocurrency Technologies: A Comprehensive Introduction”, Princeton University.
2.	Press. Antonopoulos, “Mastering Bitcoin: Unlocking Digital Cryptocurrencies”
REFERENCE BOOKS:	
1.	Satoshi Nakamoto, “Bitcoin: A Peer-to-Peer Electronic Cash System”.
2.	Dr. Gavin Wood, “ETHEREUM: A Secure Decentralized Transaction Ledger”, Yellow paper2014.
3	Nicola Atzei, Massimo Bartoletti, and Tiziana Cimoli, “A survey of attacks on Ethereum smart contracts”



Course Code	Category	L	T	P	C	I.M	E.M	Exam
B23CI4111	PE	3	--	--	3	30	70	3 Hrs.
AUGMENTED REALITY & VIRTUAL REALITY								
(For CIC)								
Pre-requisites: Image Processing, Computer Vision and Machine Vision								
Course Objectives: Students are expected								
1.	To define augmented reality (AR) and its related fields, which include virtual reality (VR) and mixed reality (MR). Analyze the display requirements for AR and the technology used for tracking. Apply AR concepts to solve real-world problems, and evaluate the usability and effectiveness of AR solutions.							
2.	To provide students with knowledge and skills for developing computer vision techniques and interaction methods specifically for Augmented Reality (AR) applications.							
3.	To familiarize students with the terminology, background, human perception, and technical underpinnings of virtual reality (VR).							
4.	To comprehend the physiology of human vision and how it affects virtual reality (VR), including rendering methods, visual perception, and VR experience optimization.							
5.	To understand interaction strategies, audio rendering for immersive experiences, and motion perception in both real and virtual environments.							
Course Out Comes: Upon completing this course, student will learn:								
S. No	OUT COME							KL
1.	Analyze differences and similarities in augmented reality systems that assess environments, tackle industry challenges, and evaluate serviceability, effectiveness, user experience, latency, field of view, and display types.							K3
2.	Use markers to position AR objects. Use infrared cameras to track them. Handle objects without markers and solve outdoor AR problems. Use voice and gestures for input. Use visuals and sound for output. Focus on combining AR with real objects and meeting AR app design needs.							K3
3.	Identify VR applications, connect VR design to human sensations, depict 3D objects and changes, and utilize light, lens, and display technologies involving cameras and vision in virtual reality.							K3
4.	Understand visual pathways from cornea to cortex, implications for VR design, depth, motion, and color perception in VR, as well as rasterization, shading, ray tracing, latency, frame rates, and distortions.							K3
5.	Analyzing motion perception, managing VR physics, addressing motion mismatch, designing interaction techniques, and applying audio rendering principles.							K3
SYLLABUS								
UNIT-I (10 Hrs)	Introduction to Augmented Reality: Augmented Reality - Defining augmented reality, history of augmented reality, Examples, Related fields. Displays: Multimodal Displays, Visual Perception, Requirements and Characteristics, Spatial Display Model, Visual							

	Displays. Tracking: Tracking, Calibration, and Registration, Coordinate Systems, Characteristics of Tracking Technology, Stationary Tracking Systems, Mobile Sensors.
UNIT-II (10 Hrs)	Computer Vision for Augmented Reality: Marker Tracking, Multiple-Camera Infrared Tracking, Natural Feature Tracking by Detection, Outdoor Tracking. Interaction: Output Modalities, Input Modalities, Tangible Interfaces, Virtual User Interfaces on Real Surfaces, Augmented Paper, Multi-view Interfaces, Haptic Interaction. Software Architectures: AR Application Requirements, Software Engineering Requirements, Distributed Object Systems, Dataflow, Scene Graphs.
UNIT-III (10 Hrs)	Introduction to Virtual Reality: Defining Virtual Reality, History of VR, Human Physiology and Perception. The Geometry of Virtual Worlds: Geometric Models, Axis-Angle Representations of Rotation, Viewing Transformations. Light and Optics: Basic Behavior of Light, Lenses, Optical Aberrations, The Human Eye, Cameras, Displays.
UNIT-IV (10 Hrs)	The Physiology of Human Vision: From the Cornea to Photoreceptors, From Photoreceptors to the Visual Cortex, Eye Movements, Implications for VR. Visual Perception: Visual Perception - Perception of Depth, Perception of Motion, Perception of Color. Visual Rendering: Visual Rendering -Ray Tracing and Shading Models, Rasterization, Correcting Optical Distortions, Improving Latency and Frame Rates, Immersive Photos and Videos.
UNIT-V (10 Hrs)	Motion in Real and Virtual Worlds: Velocities and Accelerations, The Vestibular System, Physics in the Virtual World, Mismatched Motion and Vection. Interaction: Motor Programs and Remapping, Locomotion, Social Interaction. Audio: The Physics of Sound, The Physiology of Human Hearing, Auditory Perception, Auditory Rendering.
TEXTBOOK:	
1.	“Augmented Reality: Principles & Practice” by Schmalstieg, Hollerer, Pearson Education India; First edition (12 October 2016), ISBN-10: 9332578494.
2.	Virtual Reality, Steven M. LaValle, Cambridge University Press, 2016.
REFERENCE BOOKS:	
1.	“AR Game Development”, Allan Fowler, 1st Edition, Apress Publications, 2018, ISBN 978-1484236178.
2.	“Understanding Virtual Reality: Interface, Application and Design”, William R Sherman and Alan B Craig, (The Morgan Kaufmann Series in Computer Graphics)”. Morgan Kaufmann Publishers, San Francisco, CA, 2002.
3.	“Developing Virtual Reality Applications: Foundations of Effective Design”, Alan B Craig, William R Sherman and Jeffrey D Will, Morgan Kaufmann, 2009.
4.	“Designing for Mixed Reality”, Kharis O'Connell, O'Reilly Media, Inc., 2016, ISBN:9781491962381.
5.	“Theory and applications of marker-based augmented reality”, Sanni Siltanen, Julkaisija, Utgivare Publisher. 2012. ISBN 978-951-38-7449-0.

6.	“Designing Virtual Systems: The Structured Approach”, Gerard Jounghyun Kim, 2005.
E-Resources:	
1.	https://thearea.org/augmented-reality/
2.	https://www.vntana.com/blog/creating-augmented-reality-content/
3.	https://wayup.in/en/blog/augmented-reality-ar-source-of-inspiration-and-ideas-for-web-design



Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23CI4113	SEC	--	1	2	2	30	70	3 Hrs.
ETHICAL HACKING								
(For CIC)								
Course Objectives:								
1.	Render every database-based transaction safe, secure and simple.							
2.	Transform the internet security industry by infusing professionalism and a never-before-seen efficiency.							
Course Outcomes: At the end of the course students will be able to								
S.No	Outcomes							Knowledge Level
1.	Apply monitoring techniques to manage system security.							K3
2.	Apply advanced techniques to customize and control Windows operating system.							K3
3.	Analyze authentication mechanisms.							K4
4.	Apply Perl programming constructs to develop scripts for problem solving.							K3
5.	Analyze the characteristics of computer viruses to evaluate their propagation techniques.							K4
SYLLABUS								
Unit-I	Hacking Windows: BIOS Passwords, Windows Login Passwords, Changing Windows Visuals, Cleaning Your Tracks, Internet Explorer Users, Cookies, URL Address Bar, Netscape Communicator, Cookies URL History, The Registry, Baby Sitter Programs.							
Unit-II	Advanced Windows Hacking: Editing your Operating Systems by editing Explorer.exe, The Registry, The Registry Editor, Description of .reg file, Command Line Registry Arguments, Other System Files, Some Windows & DOS Tricks, Customize DOS, Clearing the CMOS without opening your PC, The Untold Windows Tips and Tricks Manual, Exiting Windows the Cool and Quick Way, Ban Shutdowns: A Trick to Play, Disabling Display of Drives in My Computer, Take Over the Screen Saver, Pop a Banner each time Windows Boots, Change the Default Locations, Secure your Desktop Icons and Settings.							
Unit-III	Getting Past the Password: Passwords: An Introduction, Password Cracking, Cracking the Windows Login Password, The Glide Code, Windows Screen Saver Password, XOR, Internet Connection Password, Sam Attacks, Cracking Unix Password Files, HTTP Basic Authentication, BIOS Passwords, Cracking Other Passwords.							
Unit-IV	The Perl Manual: The Basics, Scalars, Interacting with User by getting Input, Chomp() and Chop(), Operators, Binary Arithmetic Operators, The Exponentiation Operator(**), The Unary Arithmetic Operators, Other General Operators, Conditional Statements, Assignment Operators. The?: Operator, Loops, The While Loop, The For Loop, Arrays, THE FOR EACH LOOP: Moving through an Array, Functions Associated with Arrays, Push() and Pop(), Unshift() and Shift(), Splice(), Default Variables, \$_, @ARGV, Input							

	Output, Opening Files for Reading, Another Special Variables.
Unit-V	How does a Virus Work? What is a Virus?, Boot Sector Viruses (MBR or Master Boot Record), File or Program Viruses, Multipartite Viruses, Stealth Viruses, Polymorphic Viruses, Macro Viruses, Blocking Direct Disk Access, Recognizing Master Boot Record (MBR) Modifications, Identifying Unknown Device Drivers, How do I make my own Virus?, Macro Viruses, Using Assembly to Create your own Virus, How to Modify a Virus so Scan won't Catch it, How to Create New Virus Strains, Simple Encryption Methods.
Textbooks:	
1.	Patrick Engbreston: "The Basics of Hacking and Penetration Testing: Ethical Hacking and Penetration Testing Made Easy", 1st Edition, Syngress publication, 2011.
2.	Ankit Fadia : "Unofficial Guide to Ethical Hacking", 3rd Edition , McMillan India Ltd, 2006.
Reference Books:	
1.	Simpson/backman/corley, "HandsOn Ethical Hacking & Network Defense International", 2nd Edition, Cengageint, 2011.
2.	Georgia Weidman: "Penetration Testing: A Hands-on Introduction to Hacking". 2014.



Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23MC4101	MC	2	--	--	--	30	--	--
CONSTITUTION OF INDIA								
(Common to all)								
Course Objectives:								
1.	To provide understanding of the fundamental principles, sources, and key features of the Indian Constitution.							
2.	To familiarize students with the structure and functioning of Union, State, and Local Governments in India.							
3.	To develop awareness about the roles of constitutional bodies like the Election Commission and welfare commissions in promoting democracy and social justice.							
Course Outcomes: At the end of the course, students will be able to								
S.No.	Outcome							Knowledge Level
1.	Apply the concepts of the Indian Constitution to interpret its features such as Citizenship, Preamble, Fundamental Rights and Duties, and Directive Principles of State Policy.							K3
2.	Apply the structure and functions of the Union Government to analyze the roles of executive, legislature, and judiciary in the Indian administrative system.							K3
3.	Apply the structure and functions of the State Government to interpret the roles of the Governor, Chief Minister, and State Secretariat in administration.							K3
4.	Apply the structure and functions of local administration to interpret the roles of district, municipal, and Panchayati Raj institutions in grassroots governance.							K3
5.	Apply the roles and functions of the Election Commission and various welfare commissions to interpret their significance in ensuring democratic governance and social justice.							K3
SYLLABUS								
UNIT-I (6 Hrs)	Introduction to Indian Constitution: Constitution meaning of the term, Indian Constitution - Sources and constitutional history, Features - Citizenship, Preamble, Fundamental Rights and Duties, Directive Principles of State Policy.							
UNIT-II (6 Hrs)	Union Government and its Administration Structure of the Indian Union: Federalism, Centre- State relationship, President: Role, power and position, PM and Council of ministers, Cabinet and Central Secretariat, Lok Sabha, Rajya Sabha, The Supreme Court and High Court: Powers and Functions.							

UNIT-III (6 Hrs)	State Government and its Administration Governor - Role and Position - CM and Council of ministers, State Secretariat: Organisation, Structure and Functions.
UNIT-IV (6 Hrs)	Local Administration - District's Administration Head - Role and Importance, Municipalities — Mayor and role of Elected Representative - CEO of Municipal Corporation Pachayati Raj: Functions PRI: Zilla Panchayat, Elected officials and their roles, CEO Zilla Panchayat: Block level Organizational Hierarchy - (Different departments), Village level - Role of Elected and Appointed officials - Importance of grass root democracy.
UNIT-V (6 Hrs)	Election Commission: Election Commission- Role of Chief Election Commissioner and Election Commissionerate State Election Commission: Functions of Commissions for the welfare of SC/ST/OBC and women.
Textbooks:	
1.	Durga Das Basu, Introduction to the Constitution of India, Prentice — Hall of India Pvt. Ltd..New Delhi.
2.	Subash Kashyap, Indian Constitution, National Book Trust.
Reference Books:	
1.	J.A. Siwach, Dynamics of Indian Government & Politics, Sterling Publishers.
2.	D.C. Gupta, Indian Government and Politics, S Chand.
3.	H.M.Sreevai, Constitutional Law of India, 4th edition in 3 volumes, Universal Law Publication.
4.	M.V. Pylee, Indian Constitution Durga Das Basu, Human Rights in Constitutional Law, Prentice Hall of India Pvt. Ltd.. New Delhi.
5.	Noorani, A.G., (South Asia Human Rights Documentation Centre), Challenges to Civil Right), Challenges to Civil Rights Guarantees in India, Oxford University Press 2012.
e-Resources	
1.	https://onlinecourses.swayam2.ac.in/e-learning/preview/imb26_mg151
2.	https://onlinecourses.nptel.ac.in/noc24_lw05/preview
3.	https://nptel.ac.in/courses/129106750



SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (AUTONOMOUS)

(Approved by AICTE, New Delhi, Affiliated to JNTUK, Kakinada)

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SRKR MARG, CHINA AMIRAM, BHIMAVARAM – 534204 W.G.Dt., A.P., INDIA

Regulation: R23		IV / IV - B.Tech. II - Semester							
CSE (IoT and Cyber Security including Blockchain Technology)									
COURSE STRUCTURE (With effect from 2023-24 admitted Batch onwards)									
Course Code	Course Name	Category	L	T	P	Cr	C.I.E.	S.E.E.	Total Marks
B23CI4201	Project and Internship	PR	-	-	24	12	60	140	200
TOTAL			0	0	24	12	60	140	200



SRKR

ENGINEERING COLLEGE

AUTONOMOUS

Course Code	Category	L	T	P	C	I.M	E.M	Exam
B23CI4201	PR	--	--	24	12	60	140	3Hrs
PROJECT WORK								
(For CIC)								
Pre-requisites: SE and Skills in different Technologies.								
Course Objectives: The main objectives of the course are to								
1.	To provide an opportunity to work in group on a topic / problem / experimentation							
2.	To encourage creative thinking process							
3.	To provide an opportunity to analyse and discuss the results to draw conclusions							
4.	To acquire and apply fundamental principles of planning and carrying out the work plan of the project through observations, discussions and decision-making process							
Course Out Comes: At the end of the course students will be able to								
S. No	OUT COMES							KL
1.	Identify a current problem through literature/field/case studies							K3
2.	Identify the objectives and methodology for solving the problem							K3
3.	Design and Develop technology/process for solving the problem							K4
4.	Evaluate the technology/process							K5
SYLLABUS								
*The object of Project Work is to enable the student to take up investigative study in the broad field of Computer Science Engineering, either fully theoretical/practical or involving both theoretical and practical work to be assigned by the Department on an individual basis or a group of students, under the guidance of a Supervisor. This is expected to provide a good initiation for the student(s) in R&D work. The assignment to normally include: a) Survey and study of published literature on the assigned topic. b) Working out a preliminary approach to the problem relating to the assigned topic. c) Conducting preliminary Analysis/Modelling/Simulation/Experiment/Design/ Feasibility. d) Preparing a written report on the study conducted for presentation to the department. e) Final Seminar, as oral Presentation before a departmental committee								